



Part I

Lab Assignment-7 (Introduction to Object-Oriented Programming)

Department: CSE

Subject: OOAD Lab (22CPP209)

Objective:

- i) Understanding using classes, nested types, static members, and templates and
- ii) use of type conversion operators and conversion constructors with/without explicit keywords.

Question 1. Design a Library Management System using classes, nested types, static members, and templates. Create a **Book** class with data members **title**, **ISBN** (nested type alias **BookID**), and **category** (nested enum: **FICTION**, **SCIENCE**, **HISTORY**, **TECHNOLOGY**), a static member to track total books, and a function to display book details. Next, implement a templated outer class **Library<T>** representing a library section with **sectionName** and **totalShelves** and a function to display section info. Inside it, define a nested templated class **Issue<U>** to issue books to members, storing an issue status (**ISSUED**, **RESERVED**) and a function **issueBook()** to print the issuance message. Create at least four books, display their details, create a library section, issue books to members using the inner class, and display the total number of books. This program should demonstrate data members, member functions, nested types, static members, and inner/outer templated classes.

Question 2: Create two classes, **CGPA** and **Percentage**, to represent grade values in different formats. The CGPA class should have a private member variable **cgpa** of type float, representing the cumulative grade point average. Similarly, the Percentage class should have a private member variable **percentage** of type float, representing the percentage grade. Your task is to explore the use of type conversion operators and constructors for converting one format to another. Also, observe the effect of using explicit keywords with the type conversion operator and constructor.